DSA03-Natural Language Processing

List of Lab Program

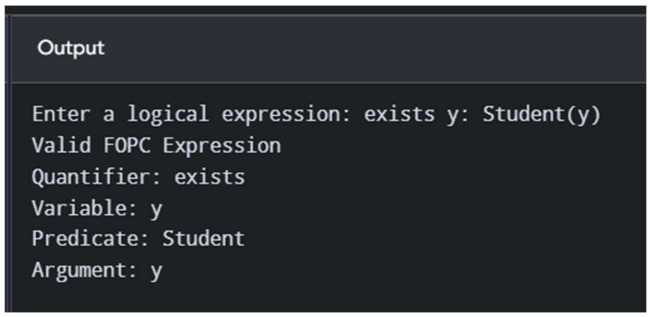
18. Implement a simple FOPC parser for basic logical expressions using python program.

Code:

```
import re

def parse_fopc(expression):
    pattern = r'^(forall|exists)\s+([a-zA-Z])\s*:\s*([A-Z][a-zA-Z]*)\(([^()]*)\)$'
    match = re.match(pattern, expression)
    if match:
        quantifier = match.group(1)
        variable = match.group(2)
        predicate = match.group(3)
        argument = match.group(4)
        print("Valid FOPC Expression")
        print("Quantifier:", quantifier)
        print("Variable:", variable)
        print("Predicate:", predicate)
        print("Argument:", argument)
    else:
        print("Invalid FOPC Expression")
expression = input("Enter a logical expression: ")
parse_fopc(expression)
```

Output:


Output
Enter a logical expression: exists y: Student(y)
Valid FOPC Expression
Quantifier: exists
Variable: y
Predicate: Student
Argument: y